

Applying Weight-of-Evidence and Information Value to Insurance Approval Models

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0. Executive Summary

1. **Project goal:** Predict approval status for travel insurance claims from nearly 57,000 records.
 - Challenge: Target is highly imbalanced — only 1.46% of claims approved.
 - Data issues: Strong skew in numerical features; high-cardinality categorical variables risk overfitting.
2. **Feature engineering:**
 - Applied Weight of Evidence (WoE) encoding to transform categorical variables into monotonic predictors.
 - Calculated Information Value (IV) for each WoE-encoded feature to select only those with strong predictive power.
 - Integrated SMOTE in the training workflow to synthesize minority-class examples and rebalance the dataset.
3. **Models tested:** Logistic Regression, Elastic Net, SVM (RBF kernel)
4. **Findings:**
 - Elastic Net and Logistic Regression balance interpretability and performance.
 - SVM overfits and is excluded from final consideration.
 - Interaction terms increase multicollinearity with little benefit.
 - Removing low-IV features slightly improves AUROC.
 - SMOTE outperforms random under-sampling in minority detection.
5. **Final tuning:** Incorporated a cost-sensitive threshold adjustment step post-prediction to minimize expected financial loss from misclassifications.
6. **Recommendation:** Deploy the Elastic Net model trained on WoE-encoded, IV-selected features with SMOTE balancing and cost-sensitive thresholding for a scalable, interpretable insurance approval decision system.

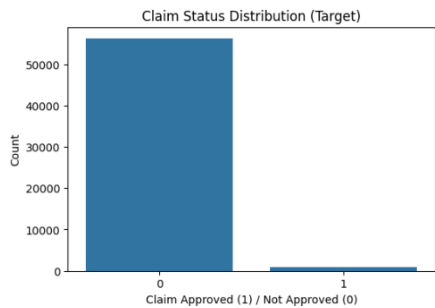
1. Introduction

This project focuses on predicting approval status for travel insurance claims. The dataset comprises over 56,000 claims, blending numerical and categorical features that capture customer demographics, transactional details, and agent information. Accurate classification is crucial for insurers seeking to balance risk management with operational efficiency.

2. Understanding the Data and Its Complexities

The dataset consists of 56,993 complete observations, with no missing values. The target variable, `claim_status`, suffers from extreme imbalance, with only about 1.46% of claims approved.

Figure 1 – Target Distribution



Numerical features exhibit pronounced irregularities. Such a skew requires careful attention to prevent models from favouring the majority class and ignoring valid claims.

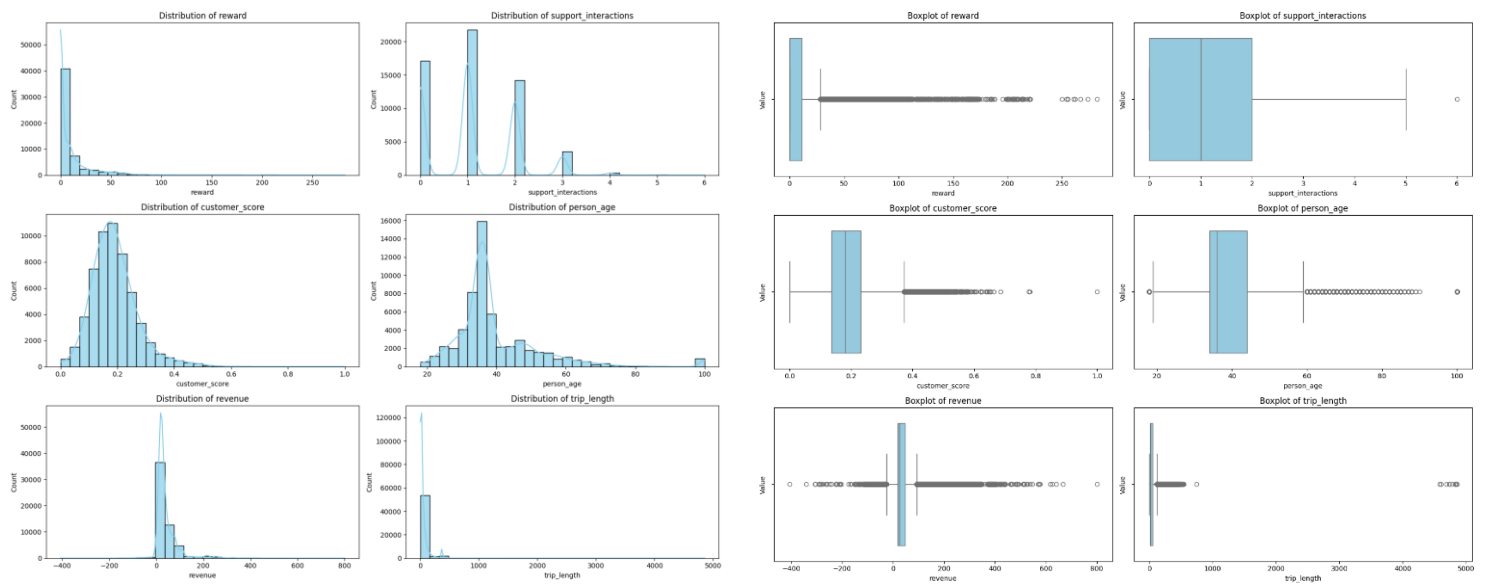
Skewness in numerical features is pronounced:

Table 1 – Skewness & Normality Calculation

Feature	Skewness
trip_length	22.32
reward	4.03
revenue	3.25
person_age	2.17
customer_score	0.95
support_interactions	0.51

The extreme skewness of `trip_length` and `reward` suggests that simple linear assumptions would be inadequate. Transformations such as logarithmic scaling are necessary to reduce distortion from outliers and improve model stability.

Figure 2 - Numerical variables exploration



Categorical variables exhibit varied complexity:

Low-cardinality variables like `person_gender`, `entity_type`, and `channel` lend themselves to straightforward encoding. In contrast, `agent_id`, `location`, and `product_id` possess high cardinality, risking model overfit if encoded naively. Effective encoding must balance granularity with parsimony.

Figure 3 - Numerical variables vs Target boxplot

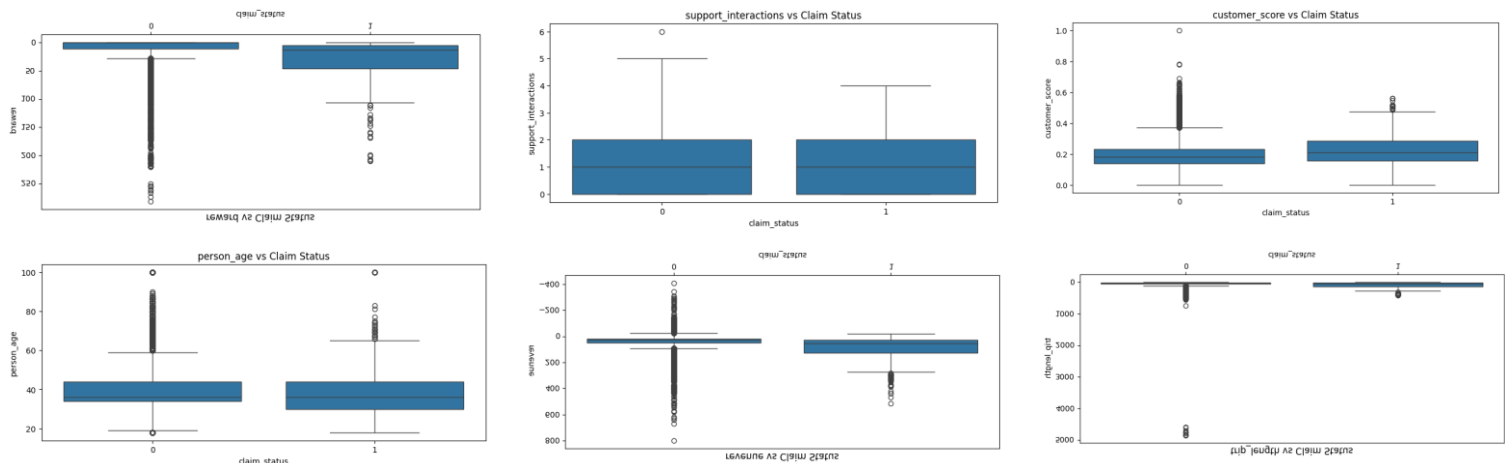
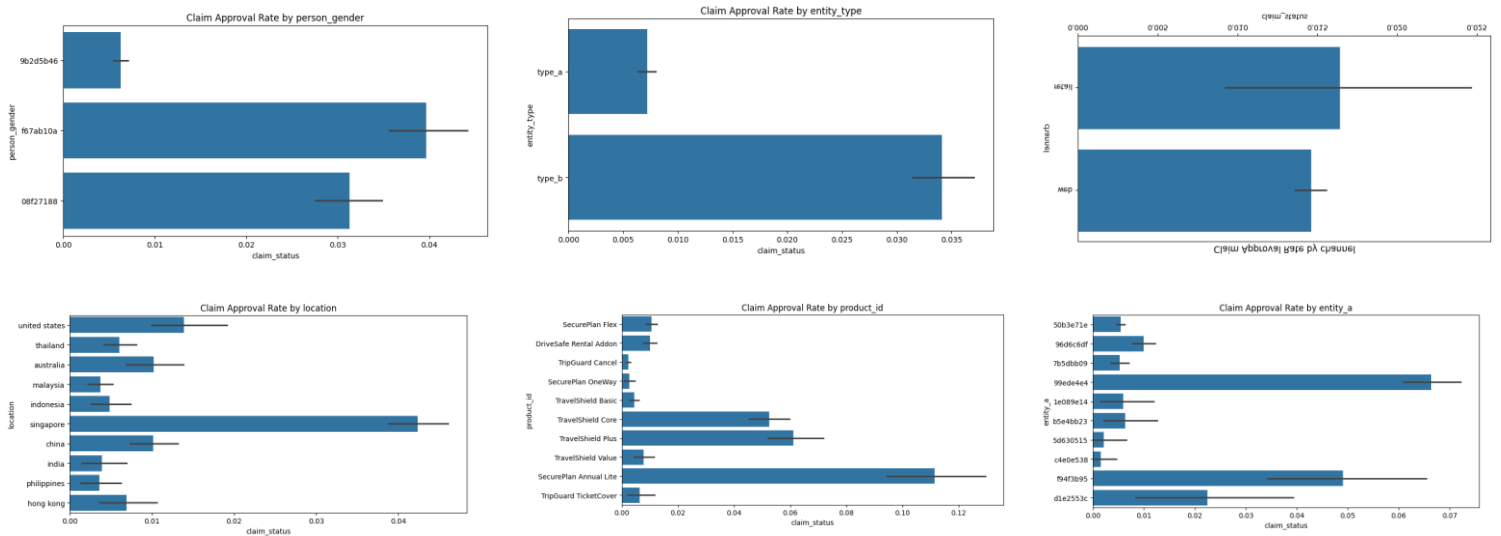


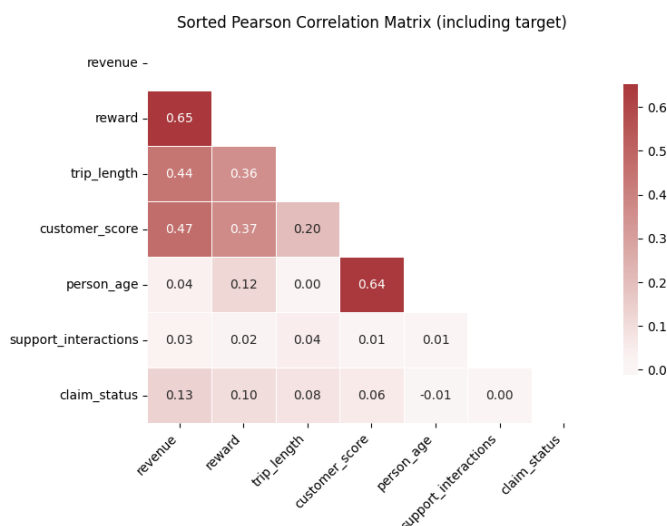
Figure 4 - Categorical vs Target variable



Correlation insights reveal generally weak associations overall, with the strongest signals arising from **reward** and **customer_score** in relation to claim approval. This indicates that predictive patterns are subtle, calling for a nuanced modelling approach.

Among categorical variables, **person_gender** and **entity_type** hold useful information, though their influence is modest and not dominant. Conversely, channel shows no meaningful relationship with claim outcomes, likely contributing noise rather than insight, and can be omitted without loss.

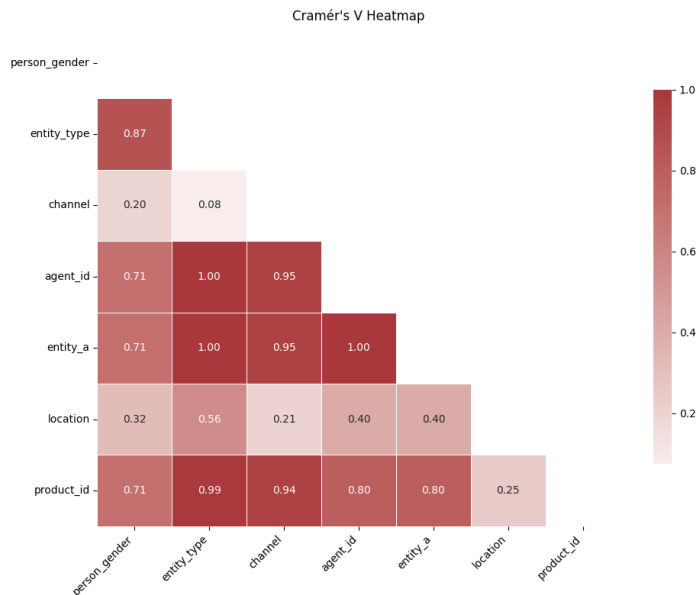
Figure 5 - Pearson Correlation Matrix of Numerical Variables x Target



More notably, strong overlap exists among **agent_id**, **entity_a**, **entity_type**, and **product_id**. This redundancy signals multicollinearity risk, necessitating dimensionality reduction or selective encoding to preserve model stability. While channel aligns closely with these variables and adds little new information, location stands apart with unique predictive value and should be retained.

Together, these findings highlight the challenge of building a model capable of managing skewed numerical distributions, high-cardinality categorical variables with complex interrelations, and severe class imbalance—all to reliably identify approved claims.

Figure 6 - Cramér's V matrix



3. Feature Engineering Strategy: Tailoring Methods to Data Realities

The data's characteristics—marked skewness, high-cardinality categorical variables, and severe class imbalance—require a modelling approach that balances complexity with interpretability and robustness.

Initial efforts applied standard transformations: logarithmic scaling to tame skewed numerical features, one-hot encoding for low-cardinality categorical variables, and frequency or target encoding for those with many levels. Basic rebalancing techniques sought to address class imbalance, while feature selection attempted to reduce dimensionality.

Yet, cross-validation revealed limited improvement. The data's subtleties—particularly among categorical variables with complex interdependencies—were not fully captured. Overlap between agent, entity, and product features complicated encoding, risking overfitting or loss of signal.

To better capture these complexities, the strategy evolved to include:

- **Weight of Evidence (WoE) Encoding:** WoE converts categorical variables into continuous scores representing their predictive relationship with claim approval. This not only reduces dimensionality but also enhances interpretability—crucial in insurance, where understanding the drivers behind decisions is essential.

- **Information Value (IV) for Feature Selection:** Paired with WoE, IV ranks variables by their predictive power, allowing weak or irrelevant predictors to be excluded early. This systematic selection reduces noise and multicollinearity, focusing the model on features with genuine explanatory value.
- **Synthetic Minority Over-sampling Technique (SMOTE):** Given the extreme class imbalance (only 1.46% of claims approved), SMOTE synthesizes new minority class examples. This yields a balanced training set, enabling the model to learn discriminative patterns more effectively without bias toward the majority.

Together, these strategies form a modelling pipeline attuned to the data's complexity and the insurance business context. Interaction terms enrich feature space with meaningful relationships; WoE and IV streamline categorical encoding and selection; SMOTE mitigates class imbalance. This synergy improves predictive accuracy and generalisability, while preserving transparency into how different features influence claim outcomes.

4. Model Selection: Balancing Interpretability, Flexibility and Performance

Given the data's complexity and the insurance business context, three models were chosen to balance interpretability, robustness, and capacity to capture non-linear relationships.

- **Logistic Regression** provides a transparent baseline. Its coefficients are directly interpretable, allowing stakeholders to understand how features influence claim approval odds. This aligns well with insurance practitioners' preference for explainable models, crucial for trust and regulatory compliance. However, standard logistic regression may struggle with high-dimensional or correlated features.
- **Elastic Net** extends logistic regression by combining L1 and L2 regularization. This helps to control multicollinearity caused by overlapping categorical variables, while simultaneously selecting relevant features. Elastic net's flexibility makes it well-suited to datasets like this one, with complex, interrelated predictors, without sacrificing interpretability.
- **Support Vector Machine with RBF kernel (SVMrbf)** introduces non-linear decision boundaries. It is well-suited for capturing subtle, complex patterns that linear models may miss, particularly when relationships between features and the target are not strictly additive or linear. While less interpretable, SVMrbf often delivers strong predictive performance, making it a useful complement to linear models.

In a nutshell, these models cover a spectrum from simplicity and transparency to complexity and flexibility. This trio allows for robust comparison, leveraging the strengths of each approach in light of the data's skewness, imbalance, and categorical intricacies.

5. Model Evaluation

5.1 Supplementary Variables Impact

Adding interaction terms between numerical features was intended to capture more complex patterns. However, this increased multicollinearity markedly: variance inflation factors rose, with some exceeding 20, signalling unstable estimates.

Table 2 - VIF Analysis

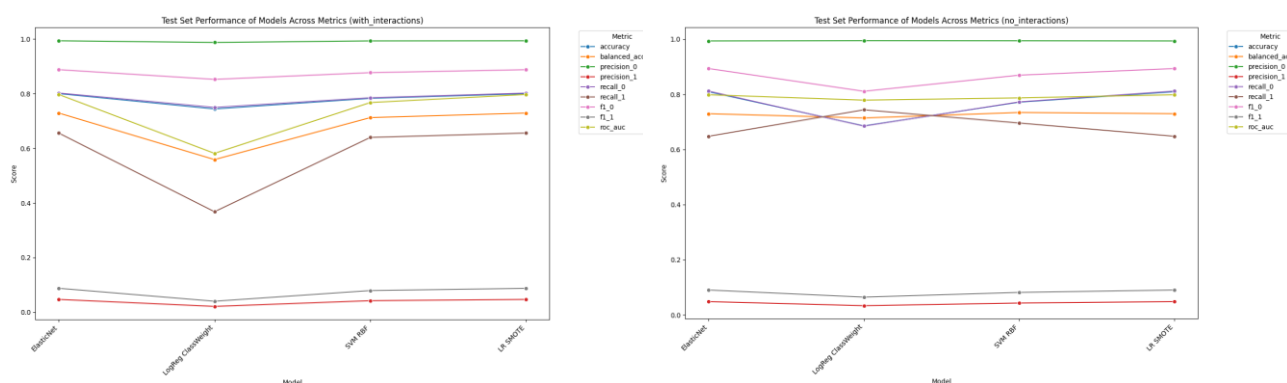
Feature	VIF (Before Interactions)	VIF (After Interactions)
customer_score	16.86	22.77
person_age	13.82	18.01
revenue	4.09	7.05
reward	2.20	8.16
support_interactions	2.17	2.83
trip_length	1.71	7.70
reward_customer	—	10.44
revenue_trip	—	5.69
age_score	—	11.20
support_trip	—	4.49

Performance on the test set showed little improvement. Elastic Net's balanced accuracy remained roughly unchanged, while logistic regression's performance declined noticeably. Support vector machines saw minor drops.

This suggests that although interaction terms can theoretically capture complex dependencies, in practice their contribution was limited or possibly offset by increased model instability due to multicollinearity.

These findings recommend caution in adding interaction features indiscriminately. The trade-off between enriched feature space and statistical stability must be carefully managed to avoid compromising predictive robustness.

Figure 7 – Interaction terms impact on model performance



5.2 Benefits of Removing Low-IV Features on Models Performance

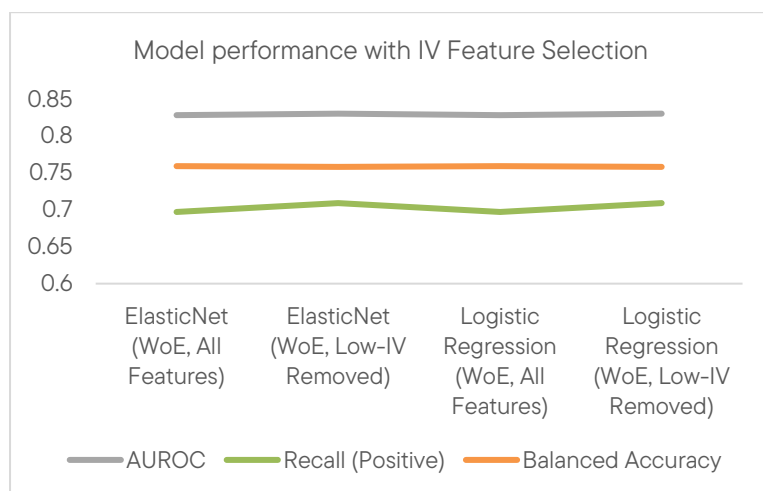
We compared the performance of ElasticNet and Logistic Regression models using WoE encoding, both with and without excluding low Information Value (IV) features. All models applied SMOTE for class imbalance correction.

The table below summarises key performance metrics averaged over validation folds:

Table 3 – Comparison of key performance metrics averaged over validation using WoE and IV

Metric	ElasticNet (WoE, All Features)	ElasticNet (WoE, Low-IV Removed)	Logistic Regression (WoE, All Features)	Logistic Regression (WoE, Low-IV Removed)
AUROC	0.828	0.83	0.828	0.83
Accuracy	0.819	0.805	0.819	0.805
Precision (Positive)	0.055	0.052	0.055	0.052
Recall (Positive)	0.697	0.709	0.697	0.709
F1 Score (Positive)	0.101	0.096	0.101	0.096
Balanced Accuracy	0.759	0.758	0.759	0.758

Figure 8 – Model performance comparison



Removing low-IV features had a modest positive impact. The model's AUROC increased slightly from approximately 0.828 to 0.830. Balanced accuracy remained stable, around 0.758 without removal and 0.757 with removal. Recall for approved claims improved marginally from about 0.697 to 0.709, while precision saw a slight decrease.

These results suggest that pruning weak predictors based on IV can simplify the model without degrading—and slightly improving—predictive performance, reinforcing IV's utility as a feature selection criterion in this context.

5.3 Rebalancing Techniques

Two rebalancing methods were compared to address class imbalance in logistic regression and ElasticNet models: **SMOTE and Random Under-Sampling**.

SMOTE consistently yielded marginally better results across all key metrics. It achieved a slightly higher AUROC (~0.83 vs. 0.83), accuracy (82% vs. 82%), and balanced accuracy (76.2% vs. 75.9%). Recall for the minority class improved with SMOTE (70.1% vs. 69.7%), as did precision and F1 scores.

Random Under-Sampling, which reduces majority class samples, performed closely but slightly below SMOTE in minority class detection metrics.

While both methods effectively address imbalance, SMOTE's synthetic minority sampling appears to enhance the model's sensitivity to approved claims, making it preferable in this context.

Table 4 - Model Performance by Rebalancing Method

Model	Rebalancing Method	AUROC	Accuracy	Precision (Positive)	Recall (Positive)	F1 Score (Positive)	Balanced Accuracy
Logistic Regression	Random Under-Sampling	0.828	0.819	0.055	0.697	0.101	0.759
Logistic Regression	SMOTE	0.830	0.823	0.056	0.701	0.103	0.762
ElasticNet	Random Under-Sampling	0.828	0.819	0.055	0.697	0.101	0.759
ElasticNet	SMOTE	0.830	0.823	0.056	0.701	0.103	0.762

5.4 Robustness Assessment with Stratified K-Fold Cross Validation

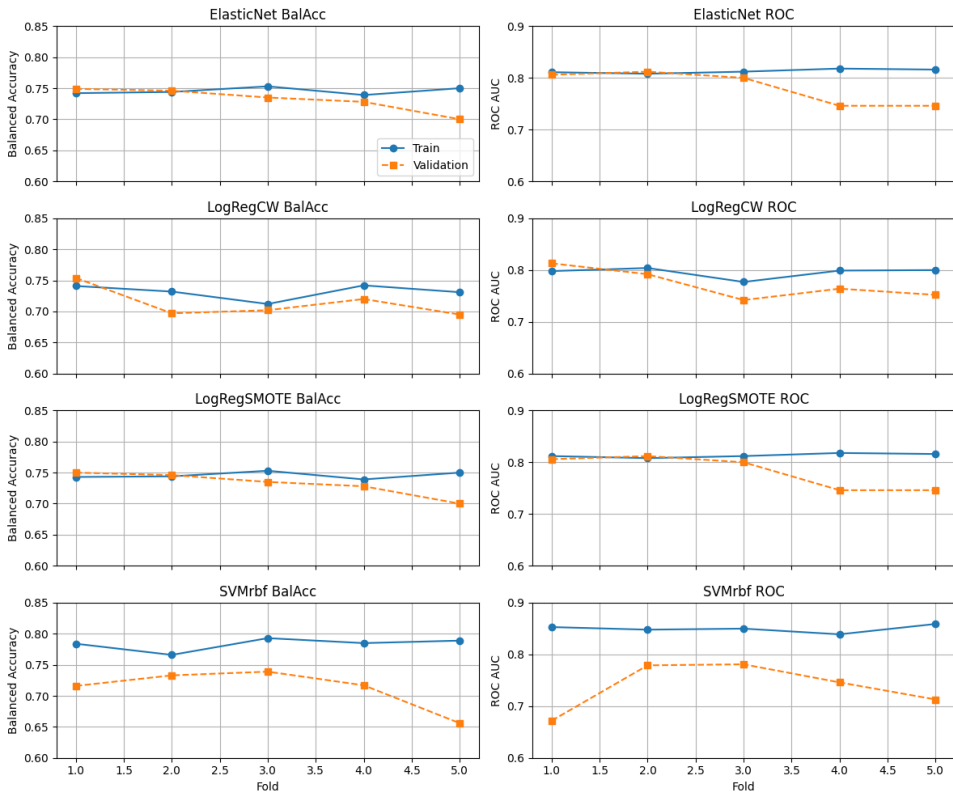
Initially, we conducted five-fold cross-validation on three candidate models: **ElasticNet, Logistic Regression with class weighting, and SVM with RBF kernel**. This preliminary assessment revealed that some models exhibited notable overfitting. In particular, the SVM with RBF kernel showed a marked disparity between training and validation performance, with ROC AUC near 0.85 on training but dropping below 0.75 on validation. Additionally, the computational demands of the SVM made extensive tuning impractical.

Table 5 - Model Performance comparison in Train & Test Set

Model	Dataset	Accuracy	Balanced Accuracy	Precision (Positive)	Recall (Positive)	F1 Score (Positive)	ROC-AUC
ElasticNet	Training	0.798	0.746	0.049	0.692	0.092	0.812

ElasticNet	Validation	0.797	0.732	0.047	0.664	0.088	0.782
Logistic Regression (Class Weight)	Training	0.768	0.731	0.046	0.694	0.086	0.796
Logistic Regression (Class Weight)	Validation	0.770	0.714	0.045	0.656	0.083	0.773
Logistic Regression (SMOTE)	Training	0.798	0.746	0.049	0.692	0.092	0.812
Logistic Regression (SMOTE)	Validation	0.798	0.732	0.047	0.664	0.088	0.782
SVM (RBF Kernel)	Training	0.769	0.783	0.049	0.798	0.092	0.850
SVM (RBF Kernel)	Validation	0.766	0.712	0.040	0.656	0.076	0.738

Figure 9 - Five fold cross validation on model performance



Consequently, we excluded the SVM from further consideration and shifted focus to a more robust evaluation framework. We employed repeated stratified K-fold cross-validation (5 folds, 3 repeats) to better estimate generalisation performance and model stability.

In this phase, we introduced Logistic Regression with Random Under-Sampling alongside ElasticNet and Logistic Regression with SMOTE, aiming to explore additional rebalancing strategies.

The results confirmed improved consistency across training and validation folds for ElasticNet and both Logistic Regression variants. Balanced accuracy and ROC AUC scores converged tightly, averaging around 0.76 and 0.83 respectively, with recall values indicating solid minority class detection.

Logistic Regression combined with Random Under-Sampling performed comparably to the SMOTE variant, with a slight trade-off in recall balanced by higher overall accuracy and computational efficiency.

This refined evaluation underscores the suitability of ElasticNet and Logistic Regression with rebalancing as reliable, efficient models ready for hyperparameter tuning and final threshold calibration.

Figure 10 - Learning Curve in ElasticNet, LR SMOTE, LR RandomUnderSampler

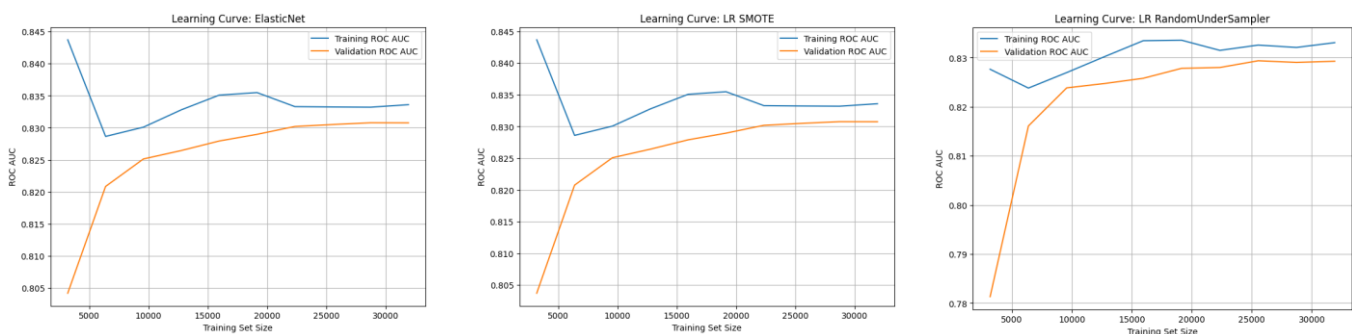


Figure 11 – Model Performance 5KFold x 3

Model	Dataset	Accuracy	Balanced Accuracy	Precision (Positive)	Recall (Positive)	F1 Score (Positive)	ROC-AUC
ElasticNet	Training	0.806	0.763	0.053	0.719	0.098	0.834
ElasticNet	Validation	0.806	0.761	0.052	0.714	0.097	0.831
Logistic Regression (SMOTE)	Training	0.806	0.763	0.053	0.719	0.098	0.834
Logistic Regression (SMOTE)	Validation	0.806	0.761	0.052	0.714	0.097	0.831
Logistic Regression (Random UnderSampler)	Training	0.812	0.761	0.054	0.708	0.100	0.833
Logistic Regression (Random UnderSampler)	Validation	0.812	0.758	0.053	0.702	0.099	0.830

Implications for Hyperparameter Tuning:

Given the close performance among logistic regression variants and ElasticNet, the next phase should focus on fine-tuning regularisation parameters (e.g., penalty strength, mixing ratio for ElasticNet) and SMOTE parameters (e.g., sampling ratio) to further optimise the trade-off between precision and recall. For SVM, tuning kernel parameters and regularisation should be considered cautiously due to computational overhead.

Cutoff Point Detection

Post-model tuning, the selection of an optimal decision threshold (cutoff) is critical to balancing false positives and false negatives in line with business objectives. Techniques such as maximizing the Youden's J statistic or cost-based thresholding will be employed to find this balance.

Final Model Selection

Based on performance, interpretability, and computational efficiency, **Logistic Regression & ElasticNet** are strong candidates for the final model. Their transparency suits regulatory needs and practical deployment, while tuning and threshold optimisation will ensure robust predictive accuracy aligned with insurer priorities.

6. Model Fine-tuning

6.1. Hyperparameter Tunning

Hyperparameter optimisation using 15-fold repeated cross-validation refined model settings for ElasticNet and logistic regression variants.

Model	Best Parameters	CV Balanced Accuracy	Notes
ElasticNet	{'clf_C': 0.1, 'clf_l1_ratio': 0.75, 'clf_penalty': 'elasticnet'}	0.761	Balanced L1/L2 regularisation, stable
Logistic Regression (SMOTE)	{'clf_C': 0.01}	0.7605	Slightly lighter regularisation, warnings
Logistic Regression (Random UnderSampler)	{'clf_C': 0.01}	0.7576	Competitive with SMOTE, simpler resampling

ElasticNet retained its optimal configuration with a balanced accuracy of approximately 0.761, utilising a moderate regularisation strength ($C=0.1$) and a predominantly L1 penalty ($l1_ratio=0.75$).

Logistic Regression with SMOTE achieved a nearly identical balanced accuracy (~ 0.7605) with a lighter regularisation ($C=0.01$). Warning messages suggest some parameters were ignored, indicating potential for further refinement.

Logistic Regression with Random Under-Sampling attained a balanced accuracy close to 0.758, also with low regularisation ($C=0.01$), confirming its competitive performance despite simpler resampling.

The consistency of balanced accuracy across these tuned models highlights their robustness. Minor differences suggest that choice among them can prioritise computational cost, interpretability, or resampling strategy.

Models and parameters have been saved for final evaluation of optimal prediction and deployment.

6.2. Threshold Tuning

a. Cost-Based Cut-off Point Selection

In the insurance claim prediction, not all classification errors are: **wrongly denying a valid claim (false negative)** risks legal exposure, reputational damage and customer loss, while **incorrectly approving a claim (false positive)** incurs financial and operational costs.

To capture this imbalance, we adopt a cost-sensitive thresholding method that weights errors by their expected financial impact. Leveraging the available revenue variable as a proxy for potential claim exposure, the total cost at a given classification threshold t is computed as:

the total cost at a given classification threshold t is computed as:

$$\text{Total Cost}(t) = \sum_{FN} \text{Cost}_{FN,i} + \sum_{FP} \text{Cost}_{FP,j}$$

Where

$$\text{Cost}_{FN,i} = C_{FN} + \text{Revenue}_i, \text{Cost}_{FP,j} = C_{FP} + \text{Revenue}_j$$

Here, C_{FN} and C_{FP} are cost multipliers reflecting the relative severity of false negatives and false positives respectively—in this study, set at 3 and 0.5. This mirrors the approach recommended in Hand (2009), which advocates cost-sensitive classification accounting for asymmetric misclassification costs in credit and insurance scoring.

By sweeping t over the unit interval, we identify the threshold minimising expected cost on a validation set, thereby aligning predictive decisions with economic priorities.

Methodology

Overall, our threshold tuning function iterates over possible classification thresholds from 0 to 1. For each threshold, it calculates the total cost incurred by false negatives and false positives:

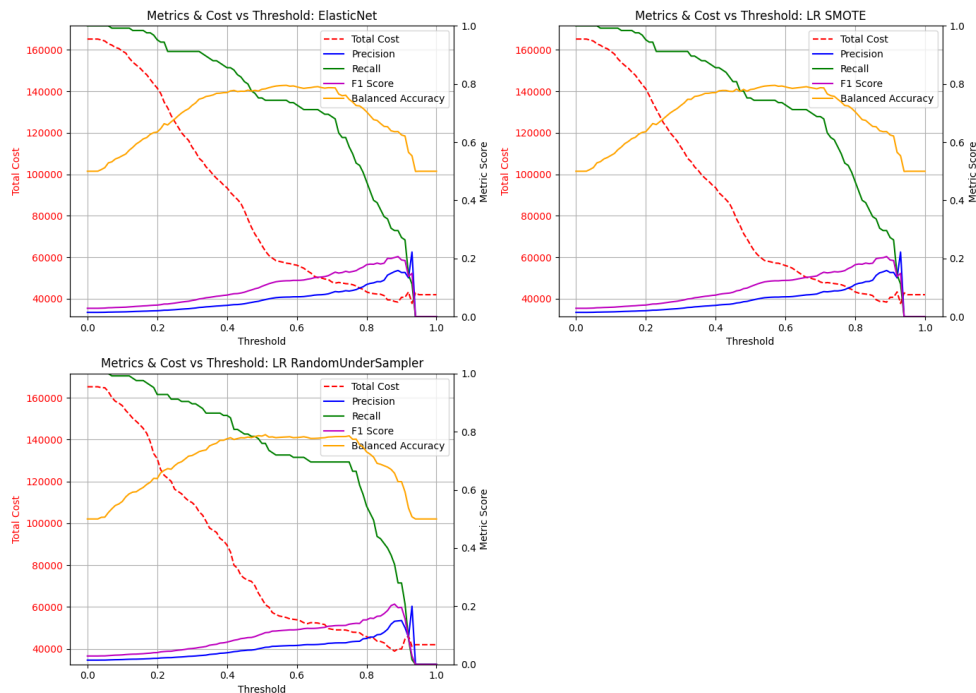
- **False negatives (FN)** are penalised at **3 times the revenue**, reflecting the high cost of missed claim approvals.
- **False positives (FP)** are penalised at **0.5 times the revenue**, representing the cost of unnecessary claim payouts or processing.

These multipliers derive from domain knowledge and industry studies, such as Hand (2009), which emphasise asymmetric misclassification costs in credit scoring and insurance.

The threshold with the minimum total cost on the validation set is selected as optimal. This threshold is then applied to the test set, evaluating its real-world financial impact alongside standard classification metrics.

Results Interpretation

Figure 12 -Threshold vs Cost Curves & Model Performance (Validation Set)



Model	Best Threshold	Validation Cost	Test Cost	Accuracy	Balanced Accuracy	Precision	Recall	F1 Score	ROC AUC
ElasticNet	0.930	37,752.91	37,676.21	0.9781	0.5160	0.0694	0.0400	0.0508	0.7980
Logistic Regression (SMOTE)	0.930	37,752.91	37,806.85	0.9780	0.5160	0.0685	0.0400	0.0505	0.7978
Logistic Regression (Random UnderSampler)	0.880	38,869.24	43,263.15	0.9577	0.5963	0.0956	0.2240	0.1340	0.7961

The optimal thresholds for ElasticNet and Logistic Regression with SMOTE converged at **0.93**, yielding a notably high classification threshold. This reflects the insurer's preference for minimising false negatives despite sacrificing some recall, resulting in **very high accuracy (~98%) but low recall (~4%)**. The balanced accuracy near 52% indicates the model favors the majority class.

Logistic Regression with Random Under-Sampling selected a lower threshold of **0.88**, with a higher recall (~22%) but increased total cost, illustrating a trade-off between sensitivity and economic efficiency.

The **cost-based total costs** highlight the practical advantage of tuning thresholds using economic considerations: **ElasticNet and Logistic Regression SMOTE** variants reduce expected financial loss more effectively than random undersampling.

b. Cut-off point by YODEN & Balanced Accuracy Maximization

To finalise the model decision, we compared three thresholding methods across our best models:

- **Cost-Based Thresholding** explicitly minimises the total financial cost by weighting false negatives and false positives according to revenue and assumed cost multipliers. This method yielded high thresholds (~0.88–0.93), resulting in excellent accuracy (~97–98%) but low recall (~4–22%). Total costs were substantially reduced compared to other methods.
- **Youden's J Statistic** maximises the difference between true positive rate and false positive rate, optimising overall classification performance without direct cost consideration. Resulting thresholds (~0.51–0.58) balance sensitivity and specificity, improving recall (~63%) but at the expense of nearly doubling total costs.
- **Balanced Accuracy Maximisation** selects the threshold that maximises balanced accuracy, balancing sensitivity to both classes equally. This method produces thresholds and performance close to Youden's, with higher recall but significantly increased cost.

The table highlights this trade-off: cost-based methods reduce expected losses substantially but favour conservative thresholds that limit approvals. Youden and balanced accuracy approaches encourage more approvals and higher recall but incur greater financial risk.

Model	Threshold	Cost	Balanced Accuracy	Precision	Recall	F1 Score	ROC AUC	Accuracy
ElasticNet-Cost-based	0.9300	37,676	0.516	0.069	0.040	0.051	0.798	0.978
ElasticNet-Youden	0.5788	59,774	0.742	0.060	0.632	0.109	0.798	0.850
ElasticNet-Balanced Accuracy Max	0.5700	60,170	0.742	0.059	0.632	0.109	0.798	0.848
LR SMOTE-Cost-based	0.9300	37,807	0.516	0.068	0.040	0.051	0.798	0.978
LR SMOTE-Youden	0.5775	59,960	0.742	0.060	0.632	0.109	0.798	0.849
LR SMOTE-Balanced Accuracy Max	0.5700	\$ 60,228	0.742	0.059	0.632	0.108	0.798	0.848

LR RDU-Cost-based	0.8800	43,263	0.596	0.096	0.224	0.134	0.796	0.958
LR RDU-Youden	0.5146	64,453	0.743	0.053	0.664	0.097	0.796	0.820
LR RDU-Balanced Accuracy Max	0.5100	64,708	0.750	0.053	0.680	0.099	0.796	0.818

A clear decision-making matrix is made for each case:

Model & Threshold	Pros	Cons	Recommended Use Case
ElasticNet / LR SMOTE (Cost-based)	Lowest total cost, very high accuracy (~97.8%)	Very low recall (~4%) — misses most positives	When minimising financial loss is the top priority and some missed claims are acceptable or managed elsewhere
ElasticNet / LR SMOTE (Youden / Balanced Accuracy)	Much higher recall (~63%), balanced accuracy (~0.74), better F1 (~0.11)	Higher total cost, lower overall accuracy (~85%)	When detecting as many positives as possible is critical, even at higher cost (e.g., fraud detection)
LR Random Under-Sampler (Cost-based)	Higher recall (~22%) than cost-based SMOTE models	Highest total cost (~43,000+), variable accuracy (~82–96%)	When catching more positives is essential and increased cost is acceptable; suitable if undersampling fits deployment
LR Random Under-Sampler (Youden / Balanced Accuracy)	Highest recall (~66–68%)	Highest total cost, lower accuracy (~82%)	Aggressive detection focus where missing positives is very costly and cost constraints are relaxed

c. Final Model Selection

ElasticNet with the Youden threshold is chosen for its balanced trade-off: it achieves substantially higher recall (~63%) and solid balanced accuracy (~74%), while maintaining reasonable costs. Its performance slightly surpasses Logistic Regression with SMOTE, offering more stable and interpretable predictions. This combination best aligns with the project's goal of balancing cost efficiency, detection capability, and model robustness.

Metric	Expected Value
Threshold	0.514589
Cost-based total cost	62578.3098
Accuracy	0.838929

Balanced Accuracy	0.740940
Precision	0.056657
Recall	0.640000
F1 Score	0.104099
ROC AUC	0.799193

7. Reflections

7.1. Challenges faced

The project confronted several significant challenges:

- **Severe class imbalance:** With approvals comprising a small fraction of the dataset, traditional modelling struggled to accurately detect minority class events without bias. Balancing sensitivity and specificity required careful rebalancing and evaluation strategies.
- **Low predictive power of many variables:** Numerous features exhibited weak or negligible association with the target, complicating feature selection. Distilling signal from noise demanded systematic approaches such as Information Value-based filtering and Weight of Evidence encoding.
- **Complexity of method combinations:** Experimenting with various rebalancing, encoding, and modelling techniques resembled cooking with a secret ingredient—finding the right mix was non-trivial. Each choice influenced others, requiring iterative tuning and deep understanding to avoid overfitting or underutilizing data.

7.2. Future work or improvement

- **Leveraging domain knowledge for scorecard development:** Integrating expert insurance knowledge could refine feature engineering, particularly in designing scorecards that translate model outputs into actionable risk assessments. This can improve interpretability and acceptance by stakeholders.

7.3. Key takeaways

- **Know your data intimately:** Understanding the data's quirks, distributions, and relationships is foundational. Blind application of algorithms without such insight risks poor performance.

- **Understand methods' strengths and limitations:** Each technique—from rebalancing to encoding to modelling—carries trade-offs. A balanced approach tailored to data characteristics and business goals is essential.

8. Reference

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